🛒 E-Commerce Sales & Category Performance Reporting
(End-to-End SQL + Excel + Power BI | YOY Commercial Analysis Project)

Executive Summary

This project replicates a real-world **weekly Category Performance Reporting System** that I owned and delivered in my previous role as a Junior Data Analyst. The reporting solution was designed to provide commercial leadership with **regular visibility into sales performance, category contribution trends, and** performance movement across major cities in India, **with deeper profitability analysis performed at a category level in Excel**.

The analysis evaluates **weekly and monthly performance across 35 cities and 13 product categories**, comparing **2023 vs 2024** to help identify growth opportunities, category momentum, and shifts in commercial performance that inform planning and inventory decisions.

The project combines **detailed Excel-based category analysis** with an **interactive Power BI executive dashboard**, demonstrating my ability to translate raw transactional data into **clear, decision-ready insights** and build reporting systems that support high-impact commercial decision-making.

Business Problem

Retail businesses operate in highly competitive environments where assortment, pricing strategy and regional performance insights are critical.

Retail leadership teams require:

- Detailed category-level performance tracking for commercial decisions
- A consolidated dashboard to monitor overall sales health, trends, and growth
- Visibility across categories, cities, and customers without manual reporting

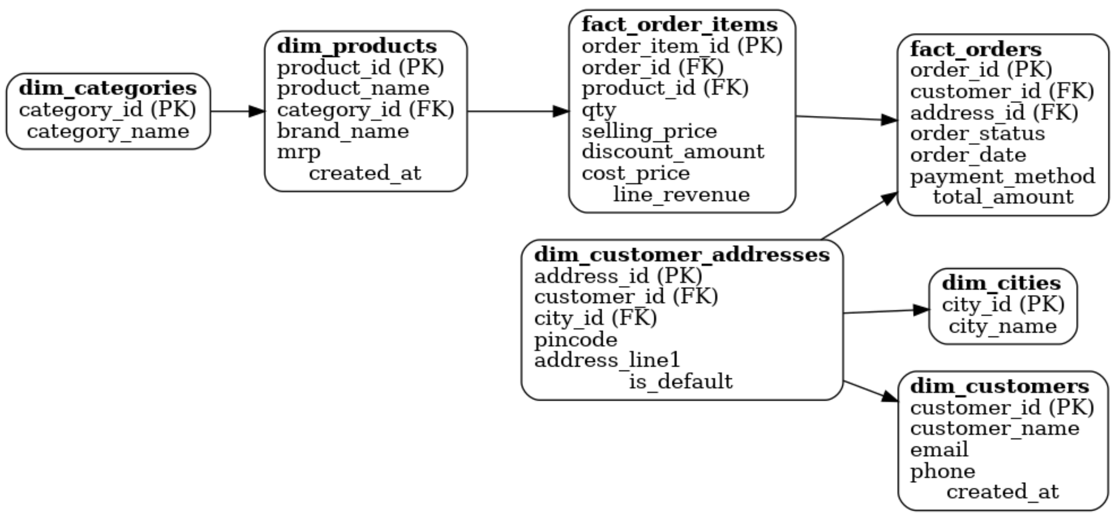
This project was designed to mirror that requirement by separating:

- **Deep-dive performance analysis**
- **High-level performance visibility**

Data Architecture & ERD

The project follows a **star-schema** design similar to real ecommerce database modeling.

Table	Description
dim_categories	Product category reference
dim_cities	City master
dim_products	Product details & category mapping
dim_customers	Customer profile
dim_customer_addresses	Address mapping to city
fact_orders	Order-level commercial information
fact_order_items	Transactional item-level details



To maintain confidentiality and avoid using any proprietary information, I created a synthetic dataset that mimics realistic ecommerce sales behavior, volume variation and seasonality.

Excel Analysis: Category Performance (Analytical Layer)

The Excel component focuses on **structured, commercial analysis**, similar to a weekly performance report prepared for leadership teams.

Analysis Performed in Excel

- Year-over-year comparison of **Sales Quantity, Net Revenue, Cost, and Profit**
- Category-level performance evaluation
- Identification of:

    - Top-performing categories
    - Declining or underperforming categories
- Contribution analysis to understand revenue concentration
- City-level aggregation to highlight key markets

Excel was used because it allows:

- Controlled calculations
- Tabular comparison
- Clear auditability of metrics
- Easy validation of commercial logic

This layer answers **“why”** performance is changing at a category and regional level.

Power BI Dashboard: Sales Overview (Executive Layer)

The Power BI dashboard was built to provide a **single-page executive overview** of overall business performance, using the same underlying sales concepts but presented at a higher level.

Key Metrics Displayed

- Total Sales (2023)
- Total Sales (2024)
- Year-over-Year Sales Growth %
- Total Orders
- Average Order Value (AOV)

These KPIs give leadership an **instant snapshot** of business performance and direction.

How Excel and Power BI Work Together (Important)

This project intentionally separates responsibilities:

Tool	Purpose
Excel	Detailed category, city, and YOY performance analysis
Power BI	High-level sales monitoring and executive visibility

In a real-world setup:

- **Excel** supports commercial reviews, planning, and investigation
- **Power BI** supports ongoing performance tracking and leadership reporting

Together, they form a **complete reporting ecosystem**, not two disconnected analyses.

Key Business Insights Enabled

- Sales declined year-over-year, requiring deeper category review
- A small number of categories drive the majority of revenue
- Revenue is concentrated in specific cities and customers
- Order volume and AOV provide additional context beyond topline sales alone
- Category-level Excel analysis helps explain trends observed in the Power BI dashboard

SQL Transformation & Metrics

1. Base CTE: order_items_enriched

The purpose of this CTE is to prepare a **clean, analysis-ready dataset** by joining all required fact and dimension tables into a single view.

In this stage, I:

- Begin with the transactional fact table **fact_order_items**, which contains one row per product per order.
- Join supporting dimensional tables to enrich the dataset:
    - **dim_products** → adds product_name and category_id
    - **dim_categories** → adds category_name
    - **fact_orders** → provides order_date to enable time-based aggregation
    - **dim_customer_addresses** → links each order to a location
    - **dim_cities** → adds the final city_name

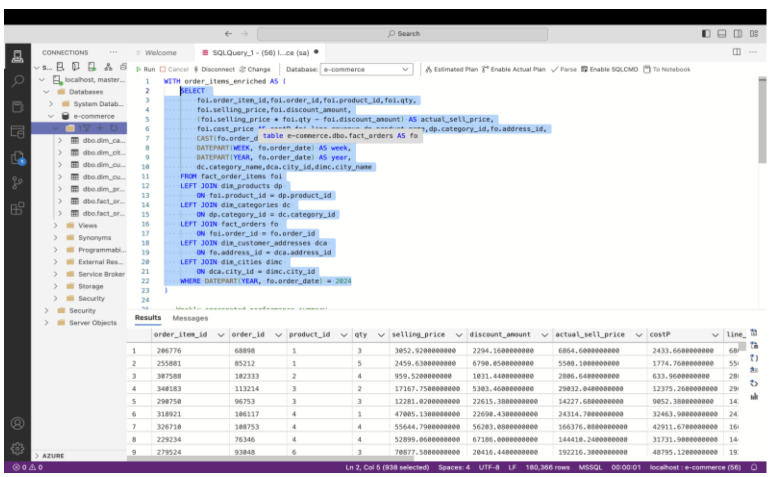
Key calculated fields

Field	Logic	Purpose
actual_sell_price	selling_price * qty – discount_amount	Net revenue after discount
costP	cost_price	Cost associated with that line item
week, year, order_date	extracted using DATEPART()	Enables weekly and yearly trend reporting

This CTE produces a **rich joined dataset** that contains:

- Category
- City
- Order timing (week and year)
- Revenue & cost at the order-item level

It serves as the **foundation** for downstream aggregation, dashboarding and YOY comparison work.



2. Weekly Aggregated Performance Summary

Based on the unified CTE dataset, the final query performs an aggregation at **Category** × **City** × **Week** level.

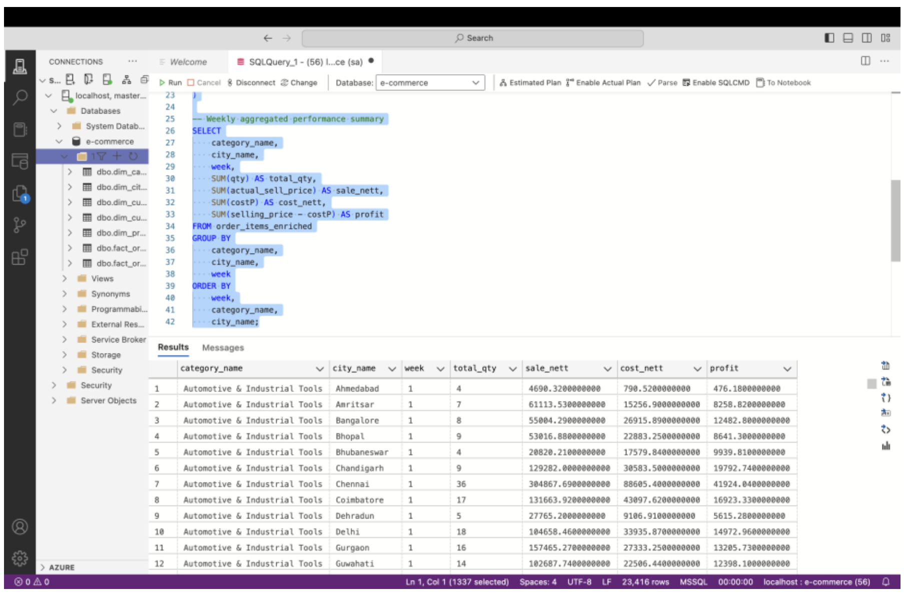
In this stage, I:

- **Group by**
  - category_name
  - city_name

    - Week
- **Aggregate key performance metrics**

Metric	Logic	Purpose
SUM(qty)	total_qty	Total units sold
SUM(actual_sell_price)	sale_nett	Total revenue after discount
SUM(costP)	cost_nett	Total cost of goods sold
SUM(selling_price - costP)	profit	Profit contribution

The result of this query is a **weekly performance table** with the below output



This is the exact structure I export to Excel to do my analysis.

Skills Demonstrated

- Commercial performance analysis
- YOY sales reporting
- Excel-based category and contribution analysis
- Power BI dashboard design & KPI storytelling
- Translating detailed analysis into executive-level insights
- Understanding of real-world reporting workflows

Dashboard Output & Supporting Analysis Performed

The main goal of the project was to conduct a detailed, multi-dimensional performance review to identify key trends, top performers, and areas for improvement.

- **Year-over-Year (YOY) Performance Analysis:** The core objective is to compare sales and profit performance metrics (**Quantity, Net Revenue, Net Cost, and Profit**) between 2024 and 2023.
- **Performance by Category:** The report classifies and ranks products by **Category** to determine which groups are driving the most value, notably via an **ABC Analysis** (A, B, C classifications).
- **Time-Series Trend Analysis:** Performance is tracked on a **weekly** basis to identify short-term trends, seasonality, or the impact of specific events across the entire business and within specific categories.
- **Geographical Performance Review:** The data is broken down by **City** and aggregated into **Areas/Regions** (e.g., North, South, East, West) to evaluate regional success and compare different markets.

Business Insights (from the excel report)

Insight	Business Impact
Mobiles & Tablets and Electronics contribute ~80% of revenue	Commercial dependency risk and major strategic focus area
Bangalore is the top performing city with highest profit contribution	Prioritize this market for promotions & premium assortment
Footwear and Beauty categories declining YOY	Opportunity for pricing or assortment optimization
ABC analysis shows 3 categories drive 85% of sales	Inventory prioritization and marketing allocation

----